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PI Name:	Shay, Jerry W. Ph.D.		
Project Title:	Risk Assessment of Space Radiation-Enhanced Colon Tumorigenesis		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Space Biology Special Category:	None		
PI Email:	jerry.shay@utsouthwestern.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	214-648-3282
Organization Name:	University of Texas Southwestern Medical Center		
PI Address 1:	Cell Biology Department		
PI Address 2:	5323 Harry Hines Blvd		
PI Web Page:			
City:	Dallas	**State:**	TX
Zip Code:	75390-7208	**Congressional District:**	30
Comments:			
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Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
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	This proposal addresses the category in the Bioastronauts Roadmap involving carcinogenesis that is associated with assessing risk of increased cancer morbidity or mortality in astronauts caused by occupational radiation exposure. Specifically we will address molecular and genetic factors that contribute to carcinogenic risk of human gastrointestinal cells (a tissue with high cellular turnover and that is particularly sensitive to the effects of radiation). We will determine the probabilities for increased carcinogenesis from space radiation that include age, latency, gender, radiation quality, dose rate, and exposure protraction. We will identify effective gastrointestinal cell biomarkers of carcinogenic risk from space radiation using both 2D and 3D cell culture systems which will be modeled and validated as well as compared to ongoing projects at our institution examining both lung and skin cells in similar models. This project should improve the estimates of the coefficients in projection models of colon cancer risk. This will include bystander effects, dose-dependent changes that may vary on genetic background, tissue microenvironment, genetic instability, and DNA repair.
Task Description:	It is well established that high energy (HZE) charged particles have a higher Relative Biological Effectiveness (RBE), as compared to X-rays. Even though HZE particle-induced cellular end points such as mutagenesis, chromosomal aberrations, and cell killing have been examined in detail, the fundamental mechanisms that lead to elevated RBE are not well understood. Furthermore, the answer to the very important question of whether exposure to HZE particles enhances cancer risk in humans remains elusive, especially in highly proliferative tissues such as the gastrointestinal system. Recent advances in the study of molecular mechanisms of DNA damage sensing, damage signaling, DNA repair, and changes in global gene expression patterns have expanded our understanding of cellular responses to low LET radiation and more recently with high LET. In this proposal, we will take advantage of these advancements to verify the hypothesis that high energy HZE particles generate a different and/or unique radiation response in mammalian cells derived from the colon as compared to X-rays. In addition, we will verify whether HZE radiation will induce a bystander effect. The proposed research should reveal the fundamental differences in cellular responses between low LET X-ray and HZE exposures in human normal colonic cells and those derived from patients with a genetic susceptibility to colon cancer (e.g. APC and HNPCC heterozygotes). These will also be compared in 2D and 3D cultures using cells derived from patients of a variety of different ages. These studies would, in turn, provide biological markers for risk assessment and guidance in radiation protection for astronauts during space missions in the future.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Colorectal cancer (CRC) is the third most prevalent cancer and third leading cause of cancer-related mortality in the United States. Data from atomic bomb survivors indicates a 70 to 80 fold risk for colon cancer development after a long-term extra-orbital space mission, such as a voyage to Mars. 2D (monolayer) and 3D cell culture models may help answer critical bio-astronautic roadmap questions, particularly in the identification of factors contributing to tissue specific probabilities of tumorigenesis after exposure to high atomic mass and energy (HZE) particles. Historically, human colonic epithelial cells have proven to be exceptionally difficult to propagate in culture. We have now determined methods to culture normal colonic epithelial cells and have created immortalized normal colonic epithelial cells able to withstand long-term cell culture and still be capable of differentiation.
	This project's overarching goal is to investigate the cellular and molecular mechanisms that contribute to the carcinogenic risk of human colonic cells and tissues after occupational radiation exposure to astronauts. Specifically, this proposal addresses the Carcinogenesis category of the Bioastronauts Roadmap. The colon represents a major target for the carcinogenic effects of radiation as it is an organ with a fast rate of cellular turnover. It also has a well characterized stem cell compartment. The precursor lesion for colorectal cancer is the adenomatous polyp and the incidence of this lesion becomes more prevalent with age (10% incidence at age 40 and 25-40% by age 60). Colorectal cancer (CRC) is the third most prevalent cancer and third leading cause of cancer related mortality in the United States (American Cancer Society, http://www.cancer.org). In 2006, it is estimated that 106,000 new cases of CRC were diagnosed, with 55,000 deaths. About 8% of CRC cases are due to familial cancer syndromes such as familial adenomatous polyposis coli (FAP) and hereditary nonpolyposis colorectal cancer (HNPCC), but the vast majority of CRCs are due to sporadic mutations in previously normal tissues. Astronauts sent on long term space missions would be presumably healthy individuals devoid of hereditary cancer syndromes or other important factors predisposing to CRC, such as inflammatory bowel disease. Irrespective, there is a very reasonable possibility for adenomas to already exist or develop in astronauts going on long-term missions, the result of which would be an increased risk for cancer development through exposure to their radiation environment. Indeed astronauts are routinely screened by colonoscopy and many have already had colonic adenomas excised. Our key objective is to determine the probabilities for increased colonic carcinogenesis from space radiation as a result of radiation dose, quality, and dose rate. Human colonic epithelial cells have proven to be exceptionally difficult to propagate in culture. Inappropriate crypt isolation techniques, inadequate cell culture substrates and nutrients, and oxidative damage have possibly contributed to the short in vitro lifespan (about ten days) of an otherwise highly replicative cell type. Our major accomplish during this pilot grant was to develop a method of sustained division of human colonic epithelial cell replication and their subsequent immortalization by introduction of cyclin-dependent kinase 4 (cdk4) and the catalytic component of telomerase (hTERT). After enzymatic processing of human colonic biopsies, isolated crypts were seeded into specially treated plastic dishes containing colonic fibroblast feeder layers, placed in a low oxygen environment (2%), and grown in a serum free environment. Cells slowly expanded from attached crypts and formed nests of epitheliod appearing cells. Feeder layers were removed with EDTA, and epithelial cells were trypsinized and seeded on to wells without feeder layers. As cells continued to replicate during this period, retroviral vectors containing cdk4 and hTERT were introduced. Uninfected cells stop dividing at 10 to 12 days of initial sub-culture and assume a flattened, enlarged appearance typical of cells stressed by inappropriate culture conditions (called premature or stress associated senescence). Cells ectopically expressing cdk4 and hTERT replicate well beyond 10 days and continue to display a cuboidal and compact appearance characteristic of healthy epithelial cells. These cells have now been in continous culture for approximately 18 months (>150 population doublings), and we continue to characterize these cells as a model for carcinogenesis studies. These cells should also be extremely useful for examining colonic cell differentiation and for cancer progression studies. The use of these cells and the successful accomplishment of the specific aims of this project should provide mechanistic dose, LET, and Z dependencies associated with the tissue microenvironment, changes in stem cell regulation, and species variations to mammalian models to achieve significant reductions in the uncertainties in risk projection for colon

Task Progress:	cancer. During this pilot project we have been studying and comparing the effects of radiation on colonic epithelial cells in monolayers, three-dimensional (3D) culture, and in intact colonic tissues. Immortalized colonic epithelial cells were irradiated at the NASA Space Radiation Laboratory (NSRL) with either 1Gy (1GeV) Fe56 or 1Gy (1GeV) protons and at various time points were stained for DNA damage responses. Results show more persisting damage at 24 hours with iron particles irradiation compared to protons. Similar results were seen in 3D colon epithelial cell cultures in which Fe56 irradiated specimens show more persisting damage at 24 hours compared to low LET gamma rays. We compared these results to colonic tissue biopsies obtained from normal human volunteers irradiated with 1 Gy gamma rays or 1Gy (1GeV) Fe56. Radiation-induced DNA damage and repair observations in low LET gamma irradiated specimens reveal more pronounced early DNA damage responses in the epithelial cell compartment compared to the stromal cell compartment. The low LET damage foci mostly disappear at 24 hours. High LET irradiated biopsy specimens show a pattern of DNA damage response predominance in the epithelial cell compartment. Un-repaired DNA damage in colonic epithelial cells and the differing repair responses between the epithelial and mesenchymal compartments in tissues may enhance tumorigenesis by both stem cell transformation and alterations in the radiation-induced permissive tissue microenvironment that may potentiate cancer progression. This initial work has been submitted for publications and is in press in Radiation Research. Summary of progress: 1. DNA damage in colonic epithelial cells is more difficult to repair when caused by iron particles compared to protons and gamma rays. 2. Iron damage persists longer in both 2D and 3D colonic epithelial cell cultures compared to protons and gamma rays. 3. Epithelial cells in intact human colonic tissues display more damage foci than surrounding stromal cells after both high and low LET radiation. All cells in intact colonic tissues are exposed to the same radiation dosage and are likely experiencing equal amounts of double strand breaks, but location in the cell cycle may explain why the replicating epithelial cells show more pronounced DNA damage response compared to quiescent stromal cells. 4. Complex DNA damage caused by high LET particles and paucity of DNA damage response in quiescent cells may lead to long standing cellular alterations (cell senescence) that can make the tissue microenvironment more permissive for tumor progression. 5. Additional cells from another patients have been established and characterized. We have an immortalized normal diploid cell line and one that is immortalized with one chromosomal change (trisomy 7). Trisomy 7 is present in about 65% of all colonic adenoma and adenocarcinoas. We have now progressed these cell partially toward tumorigenicity by reducing p53 expression (shRNA to p53) and overexpressing K-ras and will test if irradiation converts these cells to cancer forming cells. 6. We have alo passaged the cells with trisomy 7 (contain hTERT, cdk4, shRNAp53 and mutant K-ras) and observed that a subclone of these cells contain a trisomy 20 (also very common in colon cancer) and do not express APC. We are currently testing if these cells are tumorigenic and if not will subject these to low and high LET irradiation to assess risk of conversion to malignancy.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Abstracts for Journals and Proceedings	Roig AI, Shay JW. "3D Models for Risk Assessment of Space Radiation-Enhanced Colon Tumorigenesis." 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, CA, July 13 – 15, 2007. Meeting proceedings, 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, CA, July 13 – 15, 2007. , Jul-2007
Abstracts for Journals and Proceedings	Roig AI, Eskiocak U, Hight SK, Wright WE, Shay JW. "Radiation-Enhanced Cancer Progression of Human Colonic Epithelial Cells." 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 30 – July 2, 2008. Meeting proceedings, 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 30 – July 2, 2008. , Jun-2008
Articles in Peer-reviewed Journals	Roig AI, Hight SK, Shay JW. "Two- and three-dimensional models for risk assessment of radiation-enhanced colorectal tumorigenesis." Radiat Res. 2009 Jan;171(1):33-40. PMID: 19138051 (originally reported as "in press" in October 2008 report) , Jan-2009
Awards	Roig AI. "First prize poster contest, 18th Annual NASA Space Radiation Investigators' Workshop, July 2007." Jul-2007